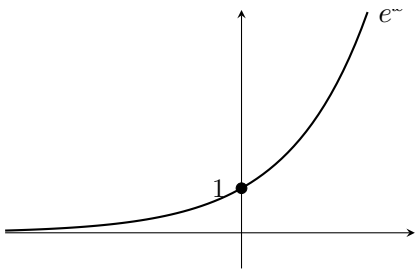
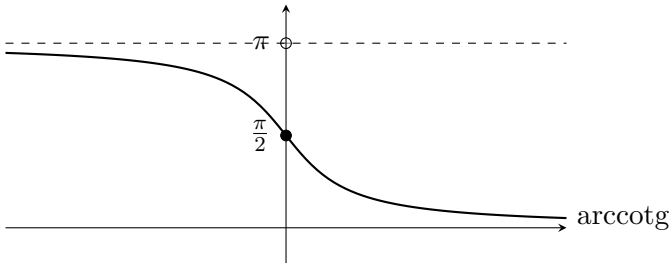
Z grafu funkce urči limity:

a) $\lim\limits_{x\to\infty} e^x = \infty$ $\lim\limits_{x\to-\infty} e^x = 0$ $\lim\limits_{x\to 0} e^x = 1$



b) $\lim\limits_{x\to\infty} \operatorname{arccotg} x = 0$ $\lim\limits_{x\to-\infty} \operatorname{arccotg} x = \pi$ $\lim\limits_{x\to 0} \operatorname{arccotg} x = \frac{\pi}{2}$



Vypočti:

a)

$$\lim_{x\to\infty} e^{\frac{2x}{x^2-1}} = e^{\lim\limits_{x\to\infty}\frac{2x}{x^2-1}} = e^{\lim\limits_{x\to\infty}\frac{2x}{x^2}} = e^{\lim\limits_{x\to\infty}\frac{2}{x}} = e^{\frac{2}{\infty}} = e^0 = \boxed{1}$$

(V čitateli i jmenovateli exponentu rozhoduje člen s nejvyšší mocninou: $2x$ a x^2.)

b)

$$\lim_{x\to\infty} \frac{3x^4-2}{2x^4+x+3} = \lim_{x\to\infty} \frac{3x^4}{2x^4} = \boxed{\frac{3}{2}}$$

(V čitateli i jmenovateli rozhoduje člen s nejvyšší mocninou: $3x^4$ a $2x^4$.)